



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Mechanical Engineering

Academic Year 2022 – 2023 (Even Semester)

Minutes of Meeting – 6th Program Assessment & Quality Improvement Committee
(PA&QIC)

Brief Introduction about PA&QIC

Objective:

To enhance the quality of the program and to ensure continuous improvement in terms of Curriculum, Teaching Learning process (TLP), Student Performance, Faculty Contributions, Infrastructural facilities, Student support system and Industry interactions by effectively involving the stakeholders

Meeting frequency : Twice per year

Roles and Responsibilities :-

Convener : Formulating the agenda and convene the meeting

Member & Representative : Share their views and suggestion for continuous improvement.
Monitor the progress of the department.

Deliverables : Submit report to Department Advisory Committee on effectiveness of program implementation in terms of each process analysis, action taken and impact.

Date of Meeting : 07.03.2023(Tuesday)

Time of Meeting : 2:30 PM to 5:00 PM

Reference : RIT/Mech./PA&QIC/06

Venue : Mechanical Seminar Hall

Members Present :

Sl. No.	Name of the Faculty Member	Designation	Responsibility
1.	Dr.S.Rajakarunakaran	Professor and Head / Mechanical	Convener
2.	Mr.M.Ashok Kumar	AP(SG) / Mechanical	Member
3.	Mr.S.Maharajan	AP(SG) / Mechanical	Member
4.	Mr.R.Prabhakaran	Assistant Professor / Mechanical	Member
5.	Mr.C.A.Yogaraja	Assistant Professor / CSE	Member
6.	Mr.A.Arun Kumar	Assistant Professor (SG) / EEE	Member
7.	Mrs.R.Ramalakshmi	Assistant Professor (SG) / ECE	Member
8.	Dr.G.Selvaraj	Assistant Professor / Maths	Member
9.	Dr.P.Suresh Kumar	Associate Professor / Mechanical	Representative

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Sl. No.	Name of the Faculty Member	Designation	Responsibility
10.	Dr.M.Lakshmanan	AP(SG)/Mechanical	Representative
11.	Dr.V.Sivakumar	Associate Professor / Mechanical	Representative
12.	Mr.J.Jerold John Britto	AP(SG) / Mechanical	Representative
13.	Mr.M.Sivagaminathan alias Balaji	Assistant Professor (SG)	Representative
14.	Mr.S.Valai Ganesh	AP(SG)/Mechanical	Representative
15.	Mr.C.Gururaj	AP(SG) / Mechanical	Representative
16.	Mr.L.Karthikeyan	AP(SG)/Mechanical	Representative
17.	Mr.G.Prabu ram	AP(SG) / Mechanical	Representative
18.	Mr.R.Arun Kumar	AP(SG) / Mechanical	Representative
19.	Mr.R.Venkatesh	AP / Mechanical	Representative
20.	Mr.M.Ramar	AP / Mechanical	Representative
21.	Mr.S. Kathiravan	AP / Mechanical	Representative

Leave / on duty:

Sl. No.	Name of the Faculty Member	Designation	PAQIC	CL/OD
1.	Mrs.R.Kalaimani	AP/Civil	Member	OD

The agenda of the meeting

1. Review of Previous meeting (5th PA&QIC)
2. New initiatives taken for quality improvement in Teaching and Learning – Outcome perspective
3. Impact analysis and effectiveness for: Quality of internal semester question paper, Assignment, Evaluation, industrial training and feedback format, mentoring system
4. Initiatives for adding Co-curricular and Extra-curricular activities towards CO attainment
5. Proposed methodologies to set target attainment for CO i) Design Domain and ii) Manufacturing Domain R 2021
6. Student's performance in Co-curricular and Extracurricular activities
7. Progress in R&D and institute industry interaction cell progress
8. Conclusion remark of the academic audit
9. Department outreach activities

Dr.V.Sivakumar, ASP/Mech. welcomed all the members of PAQIC and shared the Action Taken Report (5th PAQIC Meeting on 08.07.2022)

Activities/Tasks proposed	Action Taken
Check the Facilitation Value (FV) specified in the Assessment of Question Paper Quality template and gets the feedback from the exam cell in this regard.	• Facilitation value is included in the Form No. AC 14a / Rev.No. 03/ Effective Date: 10.02.20218
Discrimination Index (DI) of the question paper should be removed from the Assessment on Quality of Evaluation	• Discrimination Index (DI) of the question paper was removed from the Assessment on Quality of Evaluation

Activities/Tasks proposed	Action Taken
Include more responses received from parents' feedback and analysis (get the inferences and take the necessary action) of the same	Discussed and collected the opinion from the department level Criteria 1 coordinators on 18.01.2023. The analysis and action taken report to be uploaded on website before uploading AQAR 2021-22.
Include the feedback received from the class committee and course committee and any necessary action taken	
Consolidate the list of curriculum gaps identified in the previous semester and ensure that the gap is addressed through appropriate instructional methods either in the same semester or the following semester	Identified curricular gap is communicated to the AU through domain coordinators and the same is included in the course file. These are verified during the academic audit.
Give a specific certificate for those students who have completed value-added courses	MEADC2201 Advances in Manufacturing value added course certificate has been designed and submitted for approval. Financial approval received.
Assess the value added course and include CO attainment in the calculations for POs and PSOs.	Conducted end semester examinations for MEADC2201 Advances in Manufacturing. The CO attainment calculation is to be done. It is to be considered for POs and PSOs attainment calculation.
Ensure that the students are doing internships in the thrust area and enhancing their knowledge level	Students are motivated to do internship in the thrust/emerging field. It is monitored by the internship coordinator. Mr. Bala Abisek did his 6 weeks internship (August 22 – September 22) at RuTAG IIT Madras
Avoid the usage of "In-plant Training" word in the current practice.	Usage of "In-plant Training" word in the internship circular is strictly avoided.
Observe and take the necessary action on the feedback collected from the students regarding internship	Based on the positive feedback received from the previous batch students, the suggestions are given to the current batch students to choose companies. (National Small Industries Corporation)
Sort internships by i) industry, ii) start-ups, and iii) software	No. of internship done - 42 (Automobile industry – 19, Manufacturing Sector – 23)
Check the attainment value of the course C106 Engineering Graphics and double check all other courses also	Domain wise faculty members are allocated to check and consolidate the course attainment. Ref.: Department Review Meeting / 2022-23 / Even Semester / 02 / 06.03.2023
Consider for Manufacturing Domain in CO target set: Number of Times Subject handled – 10 % Faculty Experience – 10 % Course Quality – 40 % and Student Quality – 40 %	Suggestion is considered and available in 6th PAQIC presentation (Agenda 3 & 4)
Consider students' +2 PCM average for first year course outcome attainment calculation and previous semester course performance for higher semester CO attainment calculation	

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Activities/Tasks proposed	Action Taken
Assess the student's participation effectiveness in co-curricular activities through a quiz and consider it for PO attainment (20% of students' participation may be considered for attainment calculation)	
Prepare faculty wise list of participation in ATAL FDP / STTP / Webinar / etc.	Faculty wise list of participation is prepared and every Friday, Mechanical Department Activities - Student Activities, Faculty Activities, Faculty Awards and Achievements, and Event Organized are shared to the faculty members through mech@ritrjpm.ac.in
Improve the faculty interaction outside RIT	In this period, 7 faculty members acted as a resource person in 27 events.
Concentrate on the consultancy work	Concentrating consultancy work in 3D Printing and Stir Casting equipment's
Seek budget requirement from students for developing facilities which will be useful to them while doing projects.	It is proposed to collect the budget requirement from students for the FY2023-2024.
The proposed value-added course content should be in the AICTE prescribed minor courses	Proposed Value added courses - MVA023 - IoT and Augmented Reality Applications in Mechanical Engineering - MVA027 - Automation Design of Soft Robots are reflected in Model Curriculum for Minor Degree for UG Degree Courses in Engineering & Technology - 2020
Disseminate Vision /Mission / POs / PSOs /PEOs in the extended laboratories.	Disseminated Vision /Mission / POs / PSOs /PEOs in the Additive Manufacturing laboratories.
Submit the project proposals along with the student community	Faculty members along with students submitted project proposals under IE(I), TNSCST, and MSME 2.0
Review the progress of UHV/NITTTR course faculty members	- UHV Online FDP - 12 faculty members completed - UHV Refresher I FDP - 6 faculty members completed - NITTTR Course Status - 8 Modules completed by Mr.R.Prabhakaran, AP/Mech. & 7 Modules completed by Mr.R.Venkatesh, AP/Mech. (3 modules result yet to publish)
Include a resource person from industry in Extra Mural Lecture series	Mr. M. Muneeswaran, Assistant Manager, Business Development from TVS Training & Services, Chennai delivered a lecture on "Centre of Excellence - Electrical Vehicle" 03 Sep., 2022.
Include a resource person from industry in Extra Mural Lecture series	- Organized a lecture meeting on "Career Opportunities in Mechanical Engineering" at Mechanical Seminar Hall on 09 Sep., 2022. Mr. T. Thirumalai Kumar, Manager HR, Flender Drives India Private Limited, Chennai - Guest lecture "Lean in Agriculture

Activities/Tasks proposed	Action Taken
	Technology" on 21 Sep., 2022. Mr. M. Raguramsingh, Head- Process Excellence, Waycool Foods and Products Pvt., Ltd, Chennai - Organized a technology speak on "AI Driven Trends in Health Care" (Speaker: Mr. KrishnaKanth Naik, Senior Escalation Engineer, IBM India Pvt. Limited) on 27 Oct., 2022. - Organized a event on "NexGen 3D printing" on 28 Oct., 2022. A talk on New product Development by Mr. Krishna Rajagopal HTIC

Item 1: New initiatives taken for quality improvement in Teaching and Learning – Outcome perspective & Impact analysis and effectiveness for: Quality of internal semester question paper, Assignment, Evaluation, industrial training and feedback format, mentoring system
Mr.G.Prabu ram, AP(SG)Mech, and Mr.R.Arun Kumar, AP(SG)Mech, were presented Agenda 2 and 3.

- Highlighted the approval received from the Centre for academic courses, Anna University, Chennai to conduct the value added courses in the emerging areas such as i) MVA026 Python for Mechanical Engineering Systems, ii) MVA027 Automation Design of Soft Robots, iii) MVA006 Green Energy Technology and Management, and iv) MVA023 IoT and Augmented Reality Applications in Mechanical Engineering. The academic coordinator thanked all the faculty members involved in the curriculum design and development.
- Emphasized the completion of Advancement Course MEADC2201 Advances in Manufacturing by the fifty third year mechanical engineering students. This course is handled by Prof. L.Vijayaraghavan, Adjunct Professor/Mech. Engg. Number of hours handled: 50 hours (hybrid mode) Outcome/Impact of this course: Students learned the advances in manufacturing and expressed their interest to do capstone project
- Projected the initiatives taken in the analytical subjects such as CE3391 Fluid Mechanics and Machinery & ME8595 Thermal Engineering II. The subject handling faculty members were given the classroom as well as tutorial problems to their students at the beginning of the semester itself. This effort is reflected in the end semester examinations. Before revaluation, the results obtained by these two subjects are 84.38%, and 86.75% respectively. Students feel comfortable to use this study material.
- Discussed the assessment on quality of evaluation, rubrics for answer script evaluation, and inferences on assessment of question paper quality, assignment quality (Project Based Learning – ME8099 Robotics Machine Vision and Robot Programming, Quiz - CE3391 Fluid Mechanics and Machinery) and rubrics introduction for tutorials. Year wise student internship completed details along with CO attainment and mentoring system followed during this period were presented in this Agenda. The students feedback about their internship (questionnaire) has been revised and redesigned based on the POs of the NBA. Totally 50 questions with three point liker scale are used for collecting the response. To capture each PO, the questionnaire will have 3-5 questions. The questions also reflect the CDIO skills to be gained by the students. The student's response will be analyzed and insights will be used for PO attainment calculation.

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- The committee members suggested to incorporate the following in the mentoring systems based on the need arise.
 - Introduce the distinguished Alumni as one of the mentor
 - Approach the industry person and introduce them as a mentor for students seeking placement in core/software industries.

Item 2: Initiatives for adding Co-curricular and Extra-curricular activities towards CO attainment & Proposed methodologies to set target attainment for CO i) Design Domain and ii) Manufacturing Domain R 2021

Mr.R.Prabhakaran, AP/Mech. and Mr.S.Maharajan, AP/Mech. were presented Agenda 4 and 5.

In this section, existing methodology for course attainment, new approach for target setting mechanism, allocation of weightage, pre requisite for a course, course performance and student quality in previous semester for design and manufacturing domain were discussed.

Existing Methodology for course attainment – Design Domain

Subject – ME8351 Manufacturing Technology I (Regulation 2017)					
Academic Year	Grade Point Average (GPA)	GPA (out of 3 Scale)	CO target obtained	CO target (3 Scale)	Remarks
2018-2019 (Odd)	5.55	1.66	1.66	1.8	Minimum CO target of 1.8 is fixed if the calculated target value is less than 1.8
2019-2020(Odd)	5.47	1.64	CO Target $= \frac{1.66+1.64}{2} = 1.65$	1.8	
2020-2021(Odd)	7.73	2.32	CO Target $= \frac{1.66+1.64+2.32}{3} = 1.87$	1.8	
2021-2022(Odd)	8.62	2.61	CO Target = $\frac{1.66+1.64+2.32+2.61}{4} = 2.05$	1.87	
2022-2023(Odd)	-	-	-	2.05	

New Approach for Target Setting Mechanism

S.No.	Parameters	Allocated Weightage (%)
1.	Performance in Prerequisite Course	20
2.	Number of Times Subject handled	5
3.	Faculty Experience	5
4.	Course Performance	50
5.	Student Quality	20

Allocation of weightage

S.No	Parameters	Method 2 Selected	Criteria Followed
1.	Performance in Prerequisite Course, If any	20%	Considering the Grade Point Average (GPA) of Prerequisite Course Semester Result of the same batch students (Average GPA of prerequisite courses will be converted into 3 scale)

S.No	Parameters	Method 2 Selected	Criteria Followed
2.	Number of Times Subject handled	5%	Less than 2 times handled – 50% weightage (1.5) Equal to 2 times – 75% weightage (2.25) Equal to 3 and above 3 times handled – 100% weightage (3)
3.	Faculty Experience	5%	Less than 3 years experience - 50% weightage (1.5) Equal to and greater than 3 and less than 5 Years - 75% weightage (2.25) Equal to and greater than 5 Years - 100% weightage (3)
4.	Course Performance in Last four semester	50%	Average of Previous four semesters result (GPA) and convert into 3 point scale. (In case, Very New Subject – 4 th parameter can not be considered)
5.	Student Quality	20%	Average Pass Percentage of previous two semesters results (Convert into 3 Point scale)

ME3351 – Engineering Mechanics – Regulation 2021

2022-23 (Odd) (III semester) Subject Faculty: Mr.C.Gururaj

S.No.	Parameters	Allocation	Criteria Followed	CO Target obtained for each parameter
1.	Performance in Prerequisite Course, If any	20%	I Sem – MA3151 Matrices and Calculus – GPA – 4.84 I Sem – PH3151 Engineering Physics – GPA – 4.84 II Sem – MA3251 Statistics and Numerical Methods – GPA – 4.75 Convert into 3 Scale – $((4.84+4.84+4.75)/3)/10 \times 3 = 1.443$	$1.443 \times 0.2 = .288$
2.	Number of Times Subject handled	5%	Less than 2 times handled – 50% weightage (1.5) Equal to 2 times – 75% weightage (2.25) Equal to 3 and above 3 times handled – 100% weightage (3)	$1.5 \times 0.05 = 0.075$
3.	Faculty Experience	5%	Less than 3 years experience - 50% weightage (1.5) Equal to and greater than 3 and less than 5 Years - 75% weightage (2.25) Equal to and greater than 5 Years - 100% weightage (3)	$3 \times 0.05 = 0.15$
4.	Course Performance in Last four semester	50%	Average of the previous four semester results of the same course ME3351 – Engineering Mechanics $(5.512+3.365+(70\% \text{ of } (7.56+7.24))) = 4.685$ Convert into 3 Scale – 1.405	$1.405 \times 0.5 = 0.703$
5.	Student Quality in Last Two semester	20%	Overall pass percentage of past two semesters results (In case III semester subjects, consider student's performance of II and I semester) Average overall pass percentage = (64.29% &	$1.98 \times 0.2 = .396$

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S.No.	Parameters	Allocation	Criteria Followed	CO Target obtained for each parameter
			67.86%)) Converted into 3 point scale = (1.98)	
CO Target				1.612

**ME3351 – Engineering Mechanics – Regulation 2021
2022-23 (Odd) (III sem.) – Prerequisite Course**

Details	Pre requisite Course	Allocated Weightage (%)
Regulation 2021 Semester I & II (2021-22 (ODD & Even))	Students Performance: I Sem – MA3151 Matrices and Calculus – GPA – 4.84 I Sem – PH3151 Engineering Physics – GPA – 4.84 II Sem – MA3251 Statistics and Numerical Methods – GPA – 4.75 Convert into 3 Scale – $((((4.84+4.84+4.75)/3)/10)*3) = 1.443$	20% Calculation: $1.443 \times 0.2 = 0.288$

ME3351 – Engineering Mechanics (2022-23 ODD) – Course Performance in the previous semesters

Details	Course Performance			Allocated Weightage (%)
Previous Semester results of the same course: Semester III ME3351 – Engineering Mechanics	2017-2018 (EVEN)	5.512	5.512	50% Calculation: Average of GPA 4.686 3 Scale Value = 1.4057 $1.4057 \times 0.5 = 0.703$
	2018-2019 (EVEN)	3.365	3.365	
	2019-2020 (EVEN) (Covid)	7.56	70% of GPA Average Results 70 % of $(7.56+7.24)/2 = 5.18$	
	2020-2021 (EVEN) (Covid)	7.24		

ME3351 – Engineering Mechanics (2022-23 Odd) – Students Quality –previous Semester Performance

Details	Class Performance		Allocated Weightage (%)
Previous Semesters results of the Students Performance:	2021-2022 (odd)	64.29%	20% Calculation: Average 1 year Performance = 66.07 3 Scale Value = 1.98 $1.98 \times 0.2 = 0.396$
	2021-2022 (Even)	67.86%	

**ME8593 – Design of Machine Elements – Regulation 2017 - 2022-23 (Odd) (V Sem.)
Subject Faculty: Mr.R.Prabhakaran**

S.No	Parameters	Allocation	Criteria Followed	CO Target obtained for each parameter
1.	Performance in Prerequisite Course, If any	20%	IV Sem - CE8395 – Strength of Materials for Mechanical Engineers – GPA – 4.05 II Semester – GE8292 – Engineering Mechanics– GPA – 70% of 8.85 Convert into 3 Scale – $((((4.05+6.195)/2)/10)*3) = 1.536$	$1.536 \times 0.2 = 0.307$

S.No	Parameters	Allocation	Criteria Followed	CO Target obtained for each parameter
2.	Number of Times Subject handled	5%	Less than 2 times handled – 50% weightage (1.5) Equal to 2 times – 75% weightage (2.25) Equal to 3 and above 3 times handled – 100% weightage (3)	$3 \times 0.05 = 0.15$
3.	Faculty Experience	5%	Less than 3 years experience - 50% weightage (1.5) Equal to and greater than 3 and less than 5 Years - 75% weightage (2.25) Equal to and greater than 5 Years - 100% weightage (3)	$3 \times 0.05 = 0.15$
4.	Course Performance in Last four semester	50%	Average of the previous four semester results of the same course ME8593 – Design of Machine Elements $(4.265 + 5.565 + 5.78 + (70\% \text{ of } 8.83)) =$ Convert into 3 Scale – 1.634	$1.634 \times 0.5 = 0.817$
5.	Student Quality in Last Two semester	20%	Overall pass percentage of past two semesters results (In case V semester subjects, consider student's performance of IV and III semester) Average overall pass percentage = (50% & (70% of 100%)) Converted into 3 point scale = (1.8)	$1.8 \times 0.2 = 0.36$
CO Target				1.784

Proposed Methodologies to set target for CO attainment – Manufacturing Domain

Existing Methodology for course attainment

Subject – ME8351 Manufacturing Technology I (Regulation 2017)					
Academic Year	Grade Point Average (GPA)	GPA (out of 3 Scale)	CO target obtained	CO target (3 Scale)	Remarks
2018-2019 (Odd)	5.55	1.66	1.66	1.8	Minimum CO target of 1.8 is fixed if the calculated target value is less than 1.8
2019-2020 (Odd)	5.47	1.64	$\text{CO Target} = \frac{1.66 + 1.64}{2} = 1.65$	1.8	
2020-2021 (Odd)	7.73	2.32	$\text{CO Target} = \frac{1.66 + 1.64 + 2.32}{3} = 1.87$	1.8	
2021-2022 (Odd)	8.62	2.61	$\text{CO Target} = \frac{1.66 + 1.64 + 2.32 + 2.61}{4} = 2.05$	1.87	
2022-2023 (Odd)	-	-	-	2.05	

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New Approach for Target Setting Mechanism – Manufacturing Domain

S.No.	Parameters	Allocated Weightage (%)
1.	Performance in Prerequisite Course	20
2.	Number of Times Subject handled	5
3.	Faculty Experience	5
4.	Course Performance	50
5.	Student Quality	20

Allocation of weightage

S.No.	Parameters	Allocation of Weightage	Criteria Followed
1.	Performance in Prerequisite Course, If any	20%	Considering the Grade Point Average (GPA) of Prerequisite Course Semester Result of the same batch students (Average GPA of prerequisite courses will be converted into 3 scale)
2.	Number of Times Subject handled	5%	Less than 2 times handled – 50% weightage (1.5) Equal to 2 times – 75% weightage (2.25) Equal to 3 and above 3 times handled – 100% weightage (3)
3.	Faculty Experience	5%	Less than 3 years experience – 50% weightage (1.5) Equal to and greater than 3 and less than 5 Years – 75% weightage (2.25) Equal to and greater than 5 Years – 100% weightage (3)
4.	Course Performance in previous semester	50%	Average of Previous four semesters result (GPA) and convert into 3 point scale. (In case, Very New Subject – 4 th parameter can not be considered)
5.	Student Quality	20%	Average Pass Percentage of previous two semesters results (Convert into 3 Point scale)

ME3393 Manufacturing Processes – Regulation 2021- 2022-23 (Odd)

Subject Faculty: Mr.S.Maharajan

S.No	Parameters	Allocation of Weightage	Criteria Followed	CO Target obtained for each parameter
1.	Performance in Prerequisite Course, If any	20%	II Semester – PH3251 Material Science – Grade Point Average (GPA) – 6.11 Convert into 3 Scale – 1.83	$1.83 \times 0.2 = 0.37$
2.	Number of Times Subject handled	5%	Less than 2 times handled – 50% weightage (1.5) Equal to 2 times – 75% weightage (2.25) Equal to 3 and above 3 times handled – 100% weightage (3)	$3 \times 0.05 = 0.15$

S.No	Parameters	Allocation of Weightage	Criteria Followed	CO Target obtained for each parameter
3.	Faculty Experience	5%	Less than 3 years experience – 50% weightage (1.5) Equal to and greater than 3 and less than 5 Years – 75% weightage (2.25) Equal to and greater than 5 Years - 100% weightage (3)	$3 \times 0.05 = 0.15$
4.	Course Performance in Last four semester	50%	Average of the previous four semester results of the same course ME8351 Manufacturing Technology I $(5.55 + 5.47 + (70\% \text{ of } (7.73 + 8.62)/2)) = 5.58$ Convert into 3 Scale – 1.67	$1.67 \times 0.5 = 0.83$
5.	Student Quality	20%	Overall pass percentage of previous two semesters results Average overall pass percentage = $(64.29\% + 67.86\%) / 2 = 66.07\%$ Converted into 3 point scale (1.98)	$1.98 \times 0.2 = 0.40$
CO Target				1.90

ME8097 – Non Destructive Testing and Evaluation – Regulation 2017 - 2022-23 (Odd)

Subject Faculty: Mr.S.Maharajan

S.No	Parameters	Allocation	Criteria Followed	CO Target obtained for each parameter
1.	Performance in Prerequisite Course, If any	20%	ME8351 Manufacturing Technology I – GPA – 5.4 PR8592 Welding Technology – GPA – 6.8 Convert into 3 Scale – 1.83	$1.83 \times 0.2 = 0.37$
2.	Number of Times Subject handled	5%	Less than 2 times handled – 50% weightage (1.5) Equal to 2 times – 75% weightage (2.25) Equal to 3 and above 3 times handled – 100% weightage (3)	$3 \times 0.05 = 0.15$
3.	Faculty Experience	5%	Less than 3 years experience – 50% weightage (1.5) Equal to and greater than 3 and less than 5 Years – 75% weightage (2.25) Equal to and greater than 5 Years - 100% weightage (3)	$3 \times 0.05 = 0.15$
4.	Course Performance in Last four semester	50%	Average of the previous four semester results of the same course ME8097 Non Destructive Testing and Evaluation $(5.65 + 5.47 + (70\% \text{ of } (8.94 + 8.75)/2)) = 5.77$ Convert into 3 Scale (1.73)	$1.73 \times 0.5 = 0.86$

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S.No	Parameters	Allocation	Criteria Followed	CO Target obtained for each parameter
5.	Student Quality	20%	Overall pass percentage of previous two semesters results Average overall pass percentage = (59.46%) Converted into 3 point scale (1.78)	$1.78 \times 0.2 = 0.36$
CO Target				1.89

ME3393 Manufacturing Processes (2022-23 Odd) – Prerequisite Course

Details	Pre requisite Course	Allocated Weightage (%)
Regulation 2021 Semester II (2021-22 (Even))	<u>Students Performance:</u> PH3251 Material Science – Grade Point Average (GPA) – 6.11 Convert into 3 Scale – 1.83	20% Calculation: $1.83 \times 0.2 = 0.37$ Return to main

ME3393 Manufacturing Processes (2022-23 Odd) – Course Performance in previous semesters

Details	Course Performance			Allocated Weightage (%)
Previous Semester results of the same course: Semester III ME8351 Manufacturing Technology I	2018-2019 (Odd)	5.55	5.55	50% Calculation: Average of GPA 5.58 3 Scale Value = 1.67 $1.67 \times 0.5 = 0.83$
	2019-2020 (Odd)	5.47	5.47	
	2020-2021 (Odd) (Covid)	7.73	70% of GPA Average Results	
	2021-2022 (Odd) (Covid)	8.62	70 % of $(7.73 + 8.62) / 2 = 5.72$	

ME3393 Manufacturing Processes (2022-23 Odd) – Students Quality – previous Semester Performance

Details	Class Performance		Allocated Weightage (%)
Previous Semesters results of the Students Performance:	2021-2022 (Odd)	64.29%	20% Calculation: Average 1 year Performance = 66.07 3 Scale Value = 1.98 $1.98 \times 0.2 = 0.40$
	2021-2022 (Even)	67.86%	

ME8097 NDT (2022-23 Odd) VII Sem- Prerequisite Course

Details	Pre requisite Course	GPA	Allocated Weightage (%)
2020-21 (Odd) III Semester (Covid)	ME8351 Manufacturing Technology I	7.73 (70% of GPA) 5.4	20% $1.83 \times 0.2 = 0.37$
2021-22 (Even) VI Semester	PR8592 Welding Technology	6.8	
Average GPA		Average GPA - 6.1 3 Scale value = 1.83	

ME8097 NDT (2022-23 Odd) – Course Performance in last four semester

Details	Course Performance			Allocated Weightage (%)
Previous Semester results of the same course: ME8097 NTD	2018-2019 (Even) 2013 Regulation NDT	5.65	5.65	50% Calculation: Average of GPA 5.77 3 Scale Value = 1.73 $1.73 \times 0.5 = 0.86$
	2019-2020 (Odd) 2017 Regulation	5.47	5.47	
	2020-2021 (Odd) (Covid)	8.75	70% of GPA Average Results 70 % of $(8.75+8.94)/2$ $= 6.19$	
	2021-2022 (Odd) (Covid)	8.94		

ME8097 NDT (2022-23 Odd) – Students Quality – Last Two Semester Performance

Details	Class Performance		Allocated Weightage (%)
Previous Semesters results of the Students Performance:	2021-2022 (odd)	Pass Percentage 100% (COVID) Consider 70% Pass Percentage	20% Calculation: Average last two semester Performance= 59.46 3 Scale Value = 1.78 $1.78 \times 0.2 = 0.36$
	2021-2022 (Even)	Pass Percentage 48.92%	

Initiatives for adding Co-Curricular and Extra-Curricular Activities towards PO attainment

- a. Assessment methods proposed for the forthcoming academic year.

Direct	Indirect (30%)
Attainment through courses – 60% Co-curricular & Extra-curricular* – 10%	Exit Survey Program Exit Survey (10%) Alumni Exit Survey (10%) Employer Survey (10%)

*For Co-curricular activities, student's attainment will be measured based on assessment & for extra-curricular activities, students attainment will be measured based on participation (70% of the batch).

Approach for Weightage Calculations (Maximum Weightage - 10% of weightage)

S.No.	Parameters	Allocated Weightage (%)
1.	Other State Participation - Min 1 Certificate	10%
2.	International/National Level Conference Participation - Min 1 Certificate	10%
3.	Within State Participation - Min 1 Certificate (Internal)	5%
4.	Within State Participation - Min 1 Certificate (External)	5%
5.	Internship/Industrial Training	5%

II Year – 21-25 Batch Students participation Statistics

Students Count - 64

Other State Participation	Within State Participation (Inside RIT)	Within State Participation (Outside RIT)	Online Course	Internship Training	Extra Curricular Activities
7	32	24	3	10	25

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No. of Students at least attended any of the above programmes - 48 - (48/64) - 75%

No. of Students not attended any of the above programme - 16 - (16/64) - 25%

No. of students have attained (10% weightage) - 21 - 21/64 - 32.8%

No. of students have attained (5% weightage) - 27 - 27/64 - 42.15%

6. Student's performance in Co-curricular and Extracurricular activities

Mr.S.Valai Ganesh AP(SG)/Mech. presented the student participation and achievements during this period.

Outside TN Participations

- C. Mari Selvam, A. Gokul Ramanan, R. Venkadesh, S. Pradeep Durai, III Mech completed "Delivery Bike" in e-Yantra Robotics Competition organised by the IIT Bombay during 01 Oct., -13 Oct., 2022.
- R. Dinesh, Soumya Souvik Khuntia, R. Shyam Sundar, S. Jehaveerapandian, IV Mech completed "Pharma Bot" in e-Yantra Robotics Competition organised by the IIT Bombay during 01 Oct., -13 Oct., 2022.
- B. Kalirajakumar, III Mech participated in the "Diwali Fun - Poster Making Competition" organised by Daulat Ram College DRC, Delhi University DU, Delhi on 06 Nov., 2022.
- P. R. Vijay Jaganath, IV Mech attended "PLASTINDIA 2023" - The 11th International Plastics exhibition organized in pragati maidan, Delhi during 02-05 Feb., 2023.
- M. Prabhu Kumar, M. Karthikeyan, II Mech attended a webinar on "Chat GPT for coding" organised by JIS College of Engineering, West Bengal on 15 Feb., 2023.

Student Achievements

- M. Subbiah @ Selvam, R. Ganesh Kumar and R. Harish Krishnan, II Mech won "II Prize in Collage Competition" organised by Elite English Learners' Club, Ramco Institute of Technology, Rajapalayam on 02 Sep., 2022.

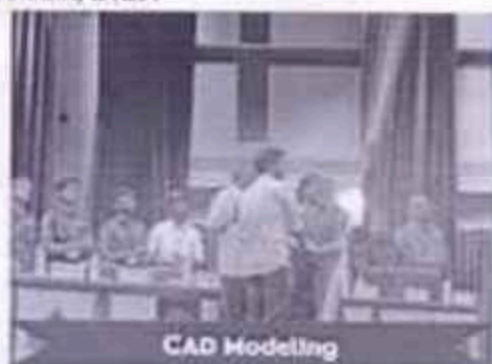


- Parthiban, III Mech won III Prize in "Debate Guru" event of Legacy'22 an intercollegiate cultural fest organized by Mepco Schlenk Engineering College, Sivakasi during 16-17 Sep., 2022.
- P. R. Vijay Jaganath, IV Mech cleared "German Goethe - Zertifikat A1 : Start Deutsch 1" and scored 82 out of 100 on 18 Sep., 2022.



- Parthiban, III Mech received "Second Prize in the Declamation event- District Level Yuva Utsav 2022" Competition held at VHNSN College, Virudhunagar on 08 October 2022
- R. Bharanidharan, K. Bharath, IV Mech received II Prize in "Mech Tech Talk event" at GenNext 2022- A National Level Technical Symposium organised by Velammal College of Engineering and Technology, Madurai on 14 Oct., 2022.
- R. Thiyanesh Sakthi, III Mech won first prize, S. Devaguru, III Mech won second prize in "New product NEXGEN Hackathon event" organised by the IIT PALS on 28 Oct., 2022.

- J. Gowtham, II Mech secured 3rd place in the "Song competition" organized by Systematic Voters Education and Electoral Participation, Election Commission of India on 16 Nov., 2022.
- P. Anandhan, V. Surya Kumar, F. Mohammed Arsath, II Mech submitted the idea for "Mechanical Palmyra tree climbing device" in Tamilnadu Agri Hackathon 2022 on 25 Nov., 2022.
- R. Dinesh, R. Shyam Sundar, IV Mech Secured 2nd Runner Up in "Autodesk Fusion 360 Challenge-Level 1" organised by the Anna University in association with Naan Mudhalvan Scheme on 05 Jan., 2023. Mr. S. Valai Ganesh, AP/Mech is the mentor for this team.
- Parthiban, III Mech awarded as "Mr. RYLA" in Rotary Youth Leadership Awards Camp held at JJI's Castle, Courtallam during 06-08 Jan., 2023.



- R. Thiyanesh Sakthi, III Mech secured First Prize in "CAD Modeling", Second Prize in "Paper Presentation" and First Prize in "Measure Hunt" events conducted by Mepco Schlenk Engineering College, Sivakasi during 17-18 Feb., 2023.
- Parthiban, III Mech won First Prize in "Paper Presentation" in ALTANZIA 2K23 Organised by National Engineering College, Kovilpatti during 03-04 Mar., 2023.
- S. Jehaveerapandian, Sowmya Souvik Khuntia received a grant of Rs.6,500/- from Tamil Nadu State Council for Science and Technology under Student Project Scheme on 03 Mar., 2023.

Student Participations- Extra- Curricular



- V. Gamel Marx, III Mech participated in the "Zonal Kabaddi Match" at Dr Sivanthi Aditanar College of Engineering on 04 Nov., 2022.
- J. Gowtham, K. Karthick, R. Ramshankar, II Mech won "Runnerup in Anna University Zone - 18 Badminton men's tournament" held at Mepco Schlenk Engineering College, Sivakasi on 06 Nov., 2022.
- D. Anithrajaseelan, IV Mech, S. K. Yuvaraj, A. Parthiban, P. Abinash, M. Sakthi Jeya Chandran, S. Devaguru III Mech, V. Palpandian, V. Harishh II Mech participated in "Anna University Zone - 18

22/11/23

Basketball men's tournament" held at National Engineering College, Kovilpatti during Co-curricular and Extra-Curricular Activities 12-13 Nov., 2022.



- S. Rahul, B. Ramkumar, B. Karthik Mahadev, V. P. Hari Haran, IV Mech, S. P. Cheran, A. Gokul Ramanan, Maruthupandi Mathavan, III Mech, participated in "Anna University Zone - 18 Football men's tournament" held at Dr Sivanthi Aditanar College of Engineering, Tiruchendur during 19-20 Nov., 2022.
- D. Anithrajaseelan, K. Rajasokanthan, M. Gowtham, K.Arul Prakash, IV Mech, S. Yuvaraj, S. Bala Chandru, III Mech, V. Palpandian, II Mech secured 4th place in "Anna University Zone - 18 Cricket men's tournament" held at Thamirabharani Engineering College, Tirunelveli during 26-28 Nov., 2022.
- V. Prakashraj, D. Gopinath, III Mech, P. Sivaprakash, II Mech participated in "Anna University Zone - 18 Hockey men's tournament" held at PSN Engineering College, Tirunelveli Co-curricular and Extra-Curricular Activities on 26 Nov., 2022.
- S. P. Cheran, S. Balachandru, A. Gokul Ramanan, M. S. Krishna kumar, M.Visva, A.C. Bharath, III MECH won runner-up award in "ISHRAE PREMIER LEAGUE" organised by ISHRAE - Madurai Chapter in Madura College Cricket Club Madurai on 28 Jan., 2023.
- D. Anith Rajaseelan, K. Arul Prakash, M. Gowtham, IV Mech, S.Balachandru, S. K. Yuvaraj, M. S. Krishnakumar, III Mech, V. Palpandian, II Mech participated in "CM Trophy- Cricket Match" conducted at Vellaichamy Nadar Polytechnic College, Virudhunagar on 21 Feb., 2023.
- J. Gowtham, K. Karthick, R. Ramshankar, II Mech participated in "CM Trophy- Badminton Match" conducted at Vellaichamy Nadar Polytechnic College, Virudhunagar on 21 Feb., 2023.
- K. Pavan Shankar, K.Karthick, R. Ram Shankar, R. Muthu Kumar, M.Karthikeyan, R. Madhavan, A. Santhosh Kumar, F. Mohamed Arsath, R. Harish Krishnan, J. Gowtham, M. Aathithyan, N. Kiruthick, K. Varun, R. Krishna, R. K. Ramnath Sarguru, R. Ramji, N. Ramji, S.Jawahar Ganesan, R. Sabarai Ravinthar, S. Sankar Ganesh, R. S. Bhuvanesh, M. Santhosh Kumar, M. Subbiah@Selvam, S. Sibi Rakesh, M. Surya Prakash, V. Palpandian, II Mech participated in "Voluntary blood donation" in service to suffering patients organized by Tamil Nadu State Aids Control Society and Tamil Nadu State Blood Transfusion Council on 21 Nov., 2022.

7. Progress in R&D and institute industry interaction cell progress

Mr.J.Jerold John Britto, AP(SG)/Mech. listed out the progress in R&D

Department of Mechanical Engineering has been recognized as a research centre with effect from 21.12.2022 for a period of three years upto December 2025 as per common renewal session based on the online inspection conducted on 01.11.2022 at 01.30 PM. Research centre recognition number: 4467817 (Ref.: Lr.No. 4678/IR/Mechanical/AR2 dated 21.12.2022)

S.No.	Supervisor Name	Recognition No.	Area of Specialization
1.	Dr. S.Rajakarunakaran	2420336	Industrial Safety Engineering, Industrial Engineering and Manufacturing Engineering
2.	Dr. P. Sureshkumar	3520021	Metallurgy, Design, Manufacturing process, Mathematical Modelling on Material processing, Severe Plastic Deformation, Composite Materials, High Entropy Alloy, Materials and Design
3.	Dr. V. Sivakumar	3020015	Thermal Energy, Solar Desalination, Renewable Energy, Internal Combustion Engineering, Heat Pipes

Publication Credentials during this period

Journal publication – 24, Scopus indexed conference – 5, and Book Chapters – 5

The list of publications during the academic year wise is disseminated to all the stakeholders through web link: https://www.ritrjpm.ac.in/images/NAAC%20DVV/AQAR_2021-22/3.1-Journal_Publication.pdf

Listed out the impact factor of the journals.

Sl. No.	Journal Name	SCI and IF
1.	Applied Soft Computing	8.263
2.	Solar Energy Materials and Solar Cells	7.305
3.	International Journal of Hydrogen Energy	7.139
4.	Journal of Materials Research and Technology	6.267
5.	Environmental Science and Pollution Research	5.19
6.	Bioinorganic Chemistry and Applications	4.724
7.	Vacuum	4.11
8.	Materials	3.748
9.	Materials Letters	3.574
10.	Environmental Progress and Sustainable Energy	2.824
11.	Knowledge and Information Systems	2.531
12.	Computing	2.42
13.	International Journal of Applied Ceramic Technology	2.328
14.	Surface Topography: Metrology and Properties	2.185
15.	Advances in Materials Science and Engineering	2.098
16.	Materials Research Express	2.025
17.	Iranian Journal of Science and Technology - Transactions of Electrical Engg.	1.89
18.	Materials Research	1.511
19.	Information Technology and Control	0.813

IPR Generations

Patent Filed- 04, FER Received- 03, and Copyright registered – 01

The patent registration and copyright particulars are periodically updated in the institute website <https://www.ritrjpm.ac.in/research/patents.php>

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Funded Projects Granted during this period

Name of the Project/ Endowments, Chairs	Name of the Principal Investigator/Co-investigator	Year of Award	Amount Sanctioned (Rs.)	Duration of the project	Name of the Funding Agency	Remarks
Experimental Evaluation of Mixed Mode Fracture Toughness of Aluminium Metal Matrix Composites and Validation Through Mathematical Modelling	Dr.S.Rajakarunakaran and Mr.C.Gururaj	2021-2022	31,970	1 Year	IEI	Fund received on April 2022
Trashbot filling alert System for smart cities	Mr.T.Selvasundar	2022-23	6500	6 Months	TNSCST	-
Smart Stick for Blind-Deaf People	Mr.M.Ramar	2022-23	7500	6 Months	TNSCST	-
Tree Measurement using LIDAR with Built-in Navigation System	Mr.S.Valai Ganesh	2022-23	7500	6 Months	TNSCST	-
Portable Rotary Fluidized bed Heat Pump Dryer for Herbal Leaves	Mr. Ashokkumar Mariarathinam	2021-22	1500000	1 year	MSME	-
Multipurpose Wheel chair	Mr. Pavithran P	2021-22	182800	1 year	MSME	-
MUSA-Holder (To protect Banana Trees from heavy winds)	Mr. SanthanaMaruthu Pandian Maruthappan	2021-22	1287500	1 year	MSME	-

Funded Project Proposal Submission Details

1. Dr.S.Rajakarunakaran (PI), Mr.J.Jerold John Britto, (Co-PI), Mr.R.Prabhakaran, Mr.R.Venkatesh, Titled on "Development of a Customized AI based FDM device with diagnostic provision for experimental studies of polymer composites for prosthetic applications with specific reference to the knee joint" DST-Device Development Program, Submitted on 15.01.2022, Amount Rs.29,01,550.
2. Dr.J.Jabinth (PI), Dr.V.Sivakumar (Co-PI), Dr.O.Senthil Kumar (Co-PI), Dr.P.Thiruramanathan (Co-PI), submitted project proposal titled as "Recirculation of the effluent from the textile industry using photocatalytic degradation treatment to achieve zero discharge in textile mill", to Optimal Water Use in Industrial Sector - 2021 - Research Stream - Establishment of Proof of Concept (POC)

- Technology Mission Division (Energy, Water & Others) – Department of Science and Technology (DST) on 29.01.2022. Amount Rs.26,26,000.

3. Dr.M.Kaliappan, (PI), Dr.S.Vimal, (CO-PI), Dr.P.Suresh kumar, (CO-PI), Dr.B.Stalin(CO-PI), Mesh, Anna University, Madurai, Titled on "Artificial Intelligence Based Baby Behavior Detection Device", DST-Device Development Program, Submitted on 14.01.2022, Amount Rs. 81,96,000.
4. Dr.S.Rajakarunakaran, Prof.and Head/Mech. submitted a project proposal titled as "Performance Study on Single/Multi-Layer Coated Slotting Machine Tool Inserts Under Cryogenic Treatment for Chain Making Machine" under the SERB Core Research Grant CRG/2022/007152/ (PAC Mechanical Manufacturing Engineering & Robotics) 1st may, 2022 & Cost: 27,10,400.00
5. Dr.P.Sureshkumar, ASP/Mech. submitted a project proposal titled as "Design and Development of Process and Material Parameter Prediction Through Intelligent Process Automation" under the SERB Core Research Grant CRG/2022/004933 (PAC Mechanical Manufacturing Engineering & Robotics)
6. Dr.V.Sivakumar, ASP/Mech. submitted a project proposal titled as "Enhancing the Functional Performance of Capacitive Deionization Units for Rural Society" under the SERB Core Research Grant CRG/2022/004932 (PAC Mechanical Manufacturing Engineering & Robotics)
7. Dr.S.Godwin Barnabas, ASP/Mech. submitted a project proposal titled as "Manufacturing of eco-friendly bumper plates from waste tyre: Techno economic feasibility study" under the SERB Core Research Grant CRG/2022/008316 (PAC Mechanical Manufacturing Engineering & Robotics)
8. Dr.J.Jabinth, AP/Mech., "Development of high strength copper hybrid metal matrix composite for wear resistant applications" under the SERB Core Research Grant CRG/2022/007112 (PAC Mechanical Manufacturing Engineering & Robotics)
9. Dr.S.Rajakarunakaran, Prof.and Head/Mech. submitted a project proposal titled as "Analysis of mechanical properties of the fabricated rubber sheet from waste tyre using Artificial Neural Network model (ANN)" under MATRICS ,File Number : MTR/2022/001014
10. Dr. P.Sureshkumar, Associate Professor, Mechanical (PI) and Dr. M. Lakshmanan, Assistant Professor (SG), Mechanical (Co-PI) submitted a project proposal titled as "Designing a workflow, performing simulations, and predicting material properties by Intelligent Process Automation, under SERB-SURE File No : SUR/2022/001912, Submitted on 19.09.2022, Amount Rs. 28,27,400.
11. Dr. P.Sureshkumar, Associate Professor, Mechanical (PI), Dr. T Manimaran, Associate Professor, Mathematics (Co-PI), Dr. P Thiruramanathan, Assistant Professor, Physics (Co-PI), Dr. E Thangam, Assistant Professor, EEE (Co-PI), and Dr. A Azhagu Jaisudhan Pazhani, Associate Professor, ECE (Co-PI) submitted a project proposal titled as "Establishment of Community Resilience Resource Centre CRRC in Rajapalayam Block, Virudhungan District, Tamil Nadu State" under DST - SEED Division File No : TPN / 85908 Submitted on 07.10.2022, Amount Rs. 28,27,400

Institute industry interaction cell progress

Mr.M.Ashok Kumar, AP(SG)/Mech. detailed the progress of IIIC.

Activity	Outcome
Students Internship	40 students from 2 nd and 3 rd year completed one week internship program – all other students will complete in this even semester. ANITH RAJASEELAN D from final year – Innowell - Intern with placement RAHUL S, MURALI KRISHNAN A, KIRUBAKARAN M from final year – Intern with project.

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Activity	Outcome
Medsby Innovation Centre	Three Day workshop conducted for II & III year RIT students from 06.10.2022 to 08.10.2022. Students Participated in RIT_IITM-PALS New Product Development Hackathon-2022 out of 100 teams our students won first prize in the title of Bicycle using Drive Shaft and Second Prize in the title of Bicycle using Kinematic Linkages . Presented a paper in NIT- Trichy in the month of August 2022. Published Four Scopus Indexed Journals.
Events organized	10 Events organized
Guest lecture delivered to Industrial persons	Mr. S. Maharajan, AP/Mech, handled the "first phase of the CNC Training programme for the employees" of Mando Automotive Private Limited, Sriperumbudur, Chennai during 26-28 Sep., 2022.
Faculty Industrial visit	Ashok Kumar M, Gururaj C and Maharajan S attended ToT programme from 02.01.2023 to 13.01.2023 organized by NIWI.

INTERNSHIP CONSOLIDATION DETAILS (01.06.2022- As on date)

Sl. No	Name of the Industry / Training Institute	Year	Duration	Name of the Student
1.	Sri Ram Karthic Polymers Pvt. Ltd.	III	01.08.2022 to 07.08.2022	Manoj Kumar A Mathan J
2.	TVS – Balasanka Motor Pvt. Ltd	II & III	01.08.2022 to 09.08.2022	V.Muneeswaran N.Ramesh Raja Keerthivasan M L Lokesh Kumar A
3.	Rajapalayam Mills Ltd	III	01.08.2022 to 09.08.2022	Gopi Krishnan M Guru Sathiesh Kumar S
4.	Sri Jayajothi Mills, Rajapalayam	III	01.08.2022 to 09.08.2022	Gopinath R and Vigneshwaran A (II Mech)
5.	JSW STEELS	II & III	01.08.2022 to 10.08.2022.	Cheran S P Karan S Sakthi Jeyachandran M Abdul Azeez S M Ahamed Sathik Ussain M Chandra Prakash G Krishna V and Vignesh S
6.	Wind Craft Energy Services, Aralvaimozhi	II	28.07.2022 to 06.08.2022.	N.Sree Vignesh
7.	Pantech Learning, Chennai	II & III	28.07.2022 to 08.08.2022.	Balachandru S , Karthik R , Mohan Raj D and Vishwa M (II Mech)

Sl. No	Name of the Industry / Training Institute	Year	Duration	Name of the Student
8.	Seven Star Aircon Ancillaries Private Limited	III	26.07.2022 to 09.08.2022	V.Krishna
9.	Ashok Leyland KSR, Rajapalayam	III	25.07.2022 to 08.08.2022	Mithran R S and Muneeswaran
10.	BSNL, Nagercoil	III	01.08.2022 to 07.08.2022	Pradeep Durai S and Venkadesh
11.	Index Control Pvt Ltd, Alangulam	III	26.07.2022 to 07.08.2022	Bharath A C , Gokul Ramanan A , Cheran S P Yuvaraj S
12.	Kalinga Stones Pvt. Ltd, Dadra and Nagar Haveli and Daman and Diu	III	26.07.2022 to 06.08.2022	Gopinath D , Rohit R, Braham Madhava Swaminath V Parthiban A
13.	Vijay Automobiles - Hero Motor Corp,	III	01.08.2022 to 10.08.2022.	V.Prakash Raj
14.	IIT MADARS - RUTAG	III	August 3rd 2022 8 th August 2022	M Bala Abshiek
15.	Salesforce Virtual Internship	III	August 10 th 2022 to 15 th August 2022	Ramkumar B
16.	3D Printing on drones, at Amrita Vishwa Vidyapeetham,	IV	17.08.2022 to 30.08.2022	Syedthowfeek S Kabil Bharathram S Dev Prasath S S

Student's Industrial visit and participation

- 20 ISHRAE Student Members attended a Technical Talk on the topic "Indian Air Conditioning Industry - A Perspective" organised by RIT ISHRAE Chapter on 17 Sep., 2022.
- R. Muthukumar, II Mech attended the webinar on "**Desiccant based dehumidification and cooling system**" organised by ISHRAE Madurai Chapter on 01 Oct., 2022.
- C. Mari Selvam, III Mech attended a webinar on "**All about Refrigerators**" organized by ISHRAE Madurai Chapter on 25 Jan., 2023.
- C. Mari Selvam, III Mech attended a webinar on "**Aircraft Air-conditioning**" organized by ISHRAE Madurai Chapter on 29 Jan., 2023.
- C. Mari Selvam, III Mech attended a webinar on "**Understanding HVAC Design from Service Perspective**" organized by ISHRAE Madurai Chapter on 09 Feb., 2023.

STUDENT PARTICIPATION-IV/INTERNSHIP

- Soumya Souvik Khuntia, S. Jehaveerapandiyan, A. Muralikrishnan, IV Mech completed "**2nd 5S Auditing**" at Gowri Metal House, Rajapalayam on 29 July, 2022.

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- S. S. Dev Prasath, Soumya Souvik Khuntia, S. Jehaveerapandiyan, A. Muralikrishnan, IV Mech completed "3rd 5S Auditing" at Gowri Metal House, Rajapalayam on 12 Aug., 2022.
- U. Ramkumar, II Mech completed a one month internship as a "Student Partner Intern at InternBug" during 30 June- 30 July, 2022.
- K. Jeya Prakash, II Mech completed a 2-month internship on "Artificial Intelligence with Python" offered by IBM, Microsoft Coimbatore and Indian Institute of Technology, Guwahati during 25 Nov., 2022- 25 Jan., 2023.
- III Year Mechanical Engineering Students visited "Autovea Metalplast, Mysore" during 23-25 Feb., 2023. Mr. J. Jerold John Britto, AP(SG)/Mech, Mr. R. Venkatesh, AP/Mech and Mr. M. Ramar, AP/Mech acted as faculty coordinators for this visit.
- II Year Mechanical Engineering Students visited "Sea Blue Shipyard Limited, Kochi" during 24-25 Feb., 2023. Dr. P. Sureshkumar, AP(SG)/Mech, Mr. M. Ashokkumar, AP(SG)/Mech acted as a faculty coordinators for this visit.

8. Conclusion remark of the academic audit

Mr.L.Karthikeyan, AP(SG)/Mech. presented the salient conclusion remark of the academic audit.

Academic Audit communication

Particulars	Remarks	Recommendations
Circular from AU AA – R 2021 Centre for Academic courses	Dated 20.01.2023 – Ref AU Clause 12	HOI – Forming committee – External expert member from near by institute AA report will be verified by AU affiliation inspection team
Nomination of AA Expert member list from dept.	Dated 07.02.2023 Ref: AU CAC – R2021	Nomination (5 expert list) proposal submitted to HOI and same is approved One expert member will be called for AA – Confidential
Circular from AC (RIT) RIT/Academic/2022-23/Even/AA/39	Dated 23.02.2023 Documents period – 2022-23 Odd sem/ Even sem	AA date – 28.02.2023 (9.30am to 4.30pm) Dept. AC – Circulate the checklist Files for R2021 – All files should be audited Audit report will be submitted to AC
Email from dept. AC AA check list	Dated 25.02.2023	Check list and course list shared to all faculty members

Academic Audit inference: R2021 files – III semester

Sl.No	Course code	Course	Specific remarks
1	ME3351	Engineering Mechanics	GATE questions may included in IAT
2	ME3391	Engineering Thermodynamics	Tutorial may included
3	CE3391	Fluid Mechanics and Machinery	GATE questions may included in IAT
4	ME3392	Engineering Materials and Metallurgy	IV may arranged

5	ME3393	Manufacturing process	IV may arranged
6	ME3381 – L	CAMD	Nil
7	ME3382 – L	MT Lab	Nil

Academic Audit inference: R2017 files – 6 Files

Sl.No.	Course code	Course	Specific remarks
1	ME8099	Robotics (PE)	Nil
2	ME8594	Dynamics of Machines	Nil
3	ME8073	Unconventional Machining process	Nil
4	MG8591	Principles of Management	Nil
5	GE8077	Total Quality Management	IV is arranged and reported in file
6	ME8711	Simulation and analysis lab	Nil

Academic Audit inferences

Audit date: 28.02.2023 (Tuesday) Type of audit: External audit (Frequency – Once in semester)		
Scope for improvement	Actions taken for Implementation	Effectiveness
Industrial visit exposure are planned in few courses which can be extended in all suitable courses for better exposure for the students. The same can be arranged in tied-up industries	Suggestions were received during 2 nd DRM / 2022-23/ Even – In all possible courses IV and activity will be planned	✓ Exposure to Industrial practice ✓ Upskilling
GATE questions are asked in few courses. The same can be given in assignments for all suitable courses	In all problematic courses, GATE questions will be included in assignments & IAT QPs	✓ Strengthen Teaching – Learning process in analytical courses ✓ Improvement in placements ✓ Improves progression towards higher studies
Measures taken for slow learners seems to be implemented in post period (eg. implemented after IAT I based on IAT I performance.).	Previous measures will be taken based on previous semester academic performance and Spcl. Classes will be planned at the beginning of classes	✓ Improvement in university results ✓ Continual improvement in academic performance & Peer learning
The target (pass percentage) set for problem as well as theory courses are same. This can be reconsidered based on the nature of the course (i.e. difficulty)	Domain coordinators are directed to work out the strategy for fixing target Extracurricular and Co-curricular activities will be included in POs attainment calculation	✓ Improvement in target & quality policy achievement ✓ Dynamic change in dept. OBE ✓ Provides 360° learning environment

22/4/23

Academic audit Inference: Positive comments

- Faculty members pursuing online courses are appreciated. Faculty members becoming expertise in interested domain by earning NPTEL domain certification was also been appreciated.
- Teaching Learning process seems to be good and the efforts taken by the faculty members were appreciated
- Infrastructure in laboratory was appreciated
- Distribution of projects under different streams was also appreciated which helps the students to choose their field of interest

9. Department Outreach Activities

Mr.C.Gururaj, AP(SG)/Mech. detailed the outreach activities done by the department faculty members and students.

Outreach activities by the Faculty members of Mechanical Engineering Department

- Dr. S. Rajakarunakaran, Prof. & Head/Mech & ISTE Executive Member participated in "118th Meeting of ISTE National Executive Council" organised by ISTE at Fairfield by Marriott Goa on 26 Aug., 2022.
- Dr. S. Godwin Barnabas, ASP/Mech, Mr. S. Valai Ganesh, AP/Mech designed and submitted an logo for the "UNESCO - INDIA - AFRICA(ULA) Hackathon" organised by MoE's Innovation Cell and AICTE on 11 Oct., 2022.



Visited to National University of Singapore

- Dr. L. Ganesan, Principal, Dr. S. Rajakarunakaran, Prof.& Head/Mech attended a meeting on "Agro Forestry" with the officials of National University of Singapore on 15 Oct., 2022.
- Mr. C. Gururaj, AP(SG)/Mech, Mr. M. Ashok Kumar, AP(SG)/Mech and Mr. S. Maharajan, AP/ Mech attended the training program "Training of Trainers for Vayumitra skill development Program" conducted by National Institute of Wind Energy (NIWE) Chennai in collaborated with TUV Rheinland - NIFE Academy during 02-13 Jan., 2023.
- In connection with the above, our institute is conducting VSDP in three job roles for the youths of Tamilnadu from 30th January 2023 onwards. The above trained faculty members train the trainees of O&M Mechanical technicians job role.
- Mr. S. Maharajan, AP/Mech attended a three-day internship training at "Mando Automotive India Private Limited, SIPCOT, Irukattukottai, Chennai" during 13-15 Dec., 2021.
- Mr. J. Jerold John Britto, AP(SG)/Mech, visited the "Ashok Leyland Hosur Plant" organised by IITM-PALS on 09 Nov., 2022.



Visited to "Mando Automotive India Private Limited - Plant 2"

- Mr. S. Maharanjan, AP/Mech, Mr. J. Jerold John Britto, AP(SG)/Mech, Mr. C. Gururaj, AP(SG)/Mech, and Mr. R. Prabhakaran, AP/Mech visited "Mando Automotive India Private Limited - Plant 2" and completed a CAD Model of Automobile Brake component during 16-17 Nov., 2022.
- Mr. S. Valai Ganesh, AP/Mech delivered a Guest Lecture on "Sensors and Arduino" organised by RIT-IIC in association with Ramco Vidayala Atal Tinkering Lab in Ramco Vidayala, Virudhunagar during 12-13 July, 2022. Mr. C. Pandimuneeswaran, LT/Mech assisted in this event.
- The faculty members of Mechanical Engineering department conducted a State level workshop "MECHRITZ - 23" for the polytechnic students of Tamilnadu in different themes.
- Dr. S. Rajakarunakaran, Prof. & Head/Mech acted as chief guest for "Arasan Dharbar-2023 National Level Technical Symposium for Polytechnic Colleges" organised by Arasan Ganesan Polytechnic College, Sivakasi on 22 Feb., 2023.
- Dr. S. Rajakarunakaran, Prof. & Head/ Mech officiated as an "Expert to do a Mock Audit- Self-Assessment Report (SAR) for the UG programmes of CSE, ECE & EEE under Tier - 1 category" invited by PSR Engineering College, Sivakasi during 24-25 Sep., 2022.
- Dr. S. Rajakarunakaran, Prof. & Head/Mech, carried out the "External Academic and Administrative Audit" in Adhiyamaan College of Engineering (Autonomous), Hosur during 12-13 Nov., 2022.
- Dr.P.Sureshkumar, ASCP/Mech conducted mock NBA audit in Shadan College of Engineering and Technology, Hyderabad during 13-14 Aug 2022.
- Dr.P.Sureshkumar, ASCP/Mech acted as resource person for orientation program of SAR Preparation for NBA in Everest Polytechnic College, Tenkasi on 04.03.2023.

Outreach activities by Mr. G.Praburam, AP(SG)/Mech.

Type of activity	Title	Institute/Organization/Forum
Guest Lecture	Ministry of Education sponsored Impact lecture on "Funding Opportunities for Early Stage Entrepreneurs".	SSM College of Pharmacy, Erode on 13 July, 2022.
Guest Lecture	"Angel Investors/ Venture capital Funding opportunity for early-stage entrepreneurs"	Institution's Innovation Council of Rajapalayam Raju's College, Rajapalayam
Participation and representation	represented RIT as IIC Innovation Ambassador in "MoE Innovation Cell and Institution's Innovation Council Regional Meet	PSNA College of Engineering & Technology, Dindigul on 08 Aug., 2022.
Guest Lecture	"Patent Searching and Drafting"	IQAC & Institution's Innovation Council of Annai Hajira Women's College, Tirunelveli on 30 Sep., 2022.

25/11/22

Type of activity	Title	Institute/Organization/Forum
Resource person	Online Webinar on "IPR Protection in Innovation and Entrepreneurship"	Department of Physics and Institution's Innovation Council of G.T.N Arts College (Autonomous), Dindigul on 25 Aug., 2022
Resource person	"Entrepreneurship Master Trainer for the three Faculty Development Programme on Entrepreneurship"	Mother Teresa Women's University in association with Entrepreneurship Development and Innovation Institute (EDII), Govt. of Tamil Nadu. held during 26-28 Sep., 2022 at the Mother Teresa Women's University research extension center, Madurai.
Resource person	"2 day Workshop on Design Thinking"	Madurai Kamaraj University in association with EDII, Govt. of Tamil Nadu during 03-04 Nov., 2022.

Our Faculty members acted as REVIEWER

- Mr. R. Arun Kumar, AP/Mech evaluated and gave suggestions for the projects done by "Ramco Private ITI students" on 14 July, 2022 and 16 July, 2022 which is to be contested in the State level competition for ITI students.
- Mr. S. Valai Ganesh, AP/Mech reviewed a paper in "Advances in Materials Science and Engineering Journal (Impact Factor= 2.098)" on 21 Sep., 2022.
- Mr. R. Arun Kumar, AP/Mech reviewed a research article titled "Effect of flow rate on the condensation characteristics of CO₂-water vapor mixture on the surface of a vertical flat plate" submitted to the Journal "Applied Thermal Engineering (IF: 6.465)" (Elsevier publication) on 12 Jan., 2023.
- Mr. S. Maharajan, AP(SG)/Mech reviewed a research article titled "Decomposition kinetics of sodium amalgam in fixed bed of WC coated tubular SS-304 packing: modelling & validation" for Asia-Pacific Journal of Chemical Engineering. WILEY Publication on 23rd DEC 2022.

Outreach activities by Students of Mechanical Engineering Department

- J. Gowtham, II Mech secured 3rd place in the "Song competition" organised by Systematic Voters Education and Electoral Participation, Election Commission of India on 16 Nov., 2022.
- P. Anandhan, V. Surya Kumar, F. Mohammed Arsath, II Mech submitted the idea for "Mechanical Palmyra tree climbing device" in Tamilnadu Agri Hackathon 2022 on 25 Nov., 2022.
- Parthiban, III Mech awarded as "Mr. RYLA" in Rotary Youth Leadership Awards Camp held at JJI's Castle, Courtallam during 06-08 Jan., 2023.
- K. Pavan Shankar, K.Karthick, R. Ram Shankar, R. Muthu Kumar, M. Karthikeyan, R. Madhavan, A. Santhosh Kumar, F. Mohamed Arsath, R. Harish Krishnan, J. Gowtham, M. Aathithyan, N. Kiruthick, K. Varun, R. Krishna, R. K. Ramnath Sarguru, R. Ramji, N. Ramji, S. Jawahar Ganesan, R. Sabarni Ravinthur, S. Sankar Ganesh, R. S. Bhuvanesh, M. Santhosh Kumar, M. Subbiah@Selvam, S. Sibi Rakesh, M. Surya Prakash, V. Palpandian, II Mech participated in "voluntary blood donation" in service to suffering patients organized by Tamil Nadu State Aids Control Society and Tamil Nadu State Blood Transfusion Council on 21 Nov., 2022.
- Soumya Souvik Khuntia, S. Jehaveerapandian, A. Muralikrishnan, IV Mech completed "SS Auditing" at Gowri Metal House, Rajapalayam on 29 July, 2022.

- P. R. Vijay Jaganath, IV Mech attended "PLASTINDIA 2023" – The 11th International Plastics exhibition organized in Pragati Maidan, Delhi during 02 – 05 Feb., 2023.

Suggestions given by the PAQIC members as follows:

- Although the term "advancement course" is already in use in R2021 practise, the members suggested coining an alternative name for the advancement course handled by the adjunct professor.
- Derive a minimum of two projects as an outcome from the course "Advances in Manufacturing Engineering."
- Tutorials, assignments and quizzes should be considered as learning and assessment tools.
- The subjects' pre-requisite tests should be divided into two categories: i) relevant/related subject content already studied, and ii) relevant/related subject content not exposed. Elective courses may not be subjected to pre-requisite testing.
- Allocation of weightage for the course Performance in the previous semester may be considered at 40% instead of 50%. The remaining 10% weightage may be given to the pre-requisite test for attainment of course.
- Recognised to consider only 10% weightage for co-curricular activities (based on the student's participation in other states, within states, or internship participation) in PO attainment.
- Set a target for individual COs / question wise targets based on the need of the course.
- Find the PEOs of the program and, based on the input from various stakeholders, change the vision and mission of the program if it requires it.
- Periodically review the short-term and long-term goals set by the department.
- Collect feedback during the conduct of any kind of active learning in the teaching and learning process.
- Do the impact analysis for the action taken for the slow learner's improvement.
- Identify the potential students from the first year itself and motivate them to secure a university rank.
- Ensure that the quality of the question paper consists of 40% in L1 and L2 and 60% in L3.
- Periodical updation required for department achievements & credentials on website.
- Appreciate the students and faculty members for their achievements.

V. Jaganath
Prepared By 26/4/23

26/4/23
NBA Coordinator/ Mech.

26/4/23
Reviewed by
(HOD/Mech.)

27/4/23
Approved by
Principal